Cyathin, a new antibiotic complex produced by *Cyathus helenae*

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The fungus *Cyathus helenae* Brodie (Gasteromycetes) produces a complex of antibiotic substances collectively called cyathin. Highest yields were obtained when the fungus was grown at room temperature for 25 days on a chemically defined liquid medium in static culture. The action spectrum of cyathin (and its components) was determined by placing cyathin-bearing paper discs on agar plates seeded with the microorganisms being tested. The cyathin complex is active against actinomycetes, bacteria (gram-positive as well as gram-negative), and some fungi, including dermatophytes. Weaker antimicrobial activity is also shown by *C. striatus* (Huds.) Willd. ex Pers., *C. poeppigii* L., and *C. limbatus* L. Seven different compounds have been isolated from the cyathin complex: 2,4,5-trihydroxybenzaldehyde ($C_7H_6O_4$), cyathin A_3($C_{20}H_{30}O_3$), cyathin A_4($C_{20}H_{30}O_4$), allocyathin A_4($C_{20}H_{30}O_4$), cyathin B_3($C_{20}H_{28}O_3$), cyathin B_4($C_{20}H_{28}O_4$), and cyathin C_5($C_{20}H_{26}O_5$). Aside from 2,4,5-trihydroxybenzaldehyde, these compounds appear not to have been previously described. A discussion of some of the structural features of cyathin A_4 and cyathin A_3 is presented

Introduction

Screening of fungi for antimicrobial substances and antitumor substances in particular has been greatly enhanced since the report by Beneke (1) of the production of calvacin by *Calvatia gigantia*. Examination of reviews of research concerning antibiotics produced by fungi (2, 3) clearly reveals that most of them are produced by microfungi (4). One of the most complete surveys devoted to the antibiotics produced by Basidiomycetes was reported by Wilkins (5, 6). With minor exceptions, almost all antimicrobial activity among 'higher' Basidiomycetes is, at present, ascribed to members of the families Polyporaceae and Agaricaceae. In his preliminary survey of the presence or absence of antibiotic activity among the Basidiomycetes, Wilkins (6) also reported tests involving three members of the gasteromycetous family Nidulariaceae, namely, *Crucibulum laeve* (Bull. ex DC.) Kambly (his *C. vulgare*), *Cyathus olla* (Batch) ex Pers., and *C. striatus* (Huds.) ex Pers. Of these, only *Cyathus striatus* exhibited any demonstrable bacteriostasis. No further investigation of the activity of the latter fungus was published by Wilkins and, apparently, the report has gone unnoticed. Brodie (7) recently described a new species of bird's nest fungus which he named *Cyathus helenae* Brodie. Olchowecki (8), studying *C. helenae* in culture, made the chance observation that the mycelium inhibited the growth of bacteria with which some culture plates had accidentally become contaminated. The antibiotic, which was named 'cyathin,' inhibited the growth of *Staphylococcus aureus*, *Micrococcus* sp., *Escherichia coli*, and *Serratia* sp. Of other species of the genus *Cyathus* screened for antibiotic activity, some indication of positive reaction was reported by Olchowecki for *C. striatus*, *C. limbatus* Tul., and *C. poeppigii* Tul. (8).

We undertook a detailed study of the biological activity and chemical characterization of the active components of cyathin, and the results are reported herein.

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Materials and Methods

Cultures of various species of *Cyathus* were obtained from the collections of Prof. H. J. Brodie. *Cyathus helenae* strain 1500-102 (haploid), was used chiefly throughout this investigtion. The characteristics of this fungus and its mating reactions have been described by Brodie (7) and Olchowecki and Brodie (9). Cultures of other fungi were obtained from Drs. L. L. Kennedy and J. W. Carmichael. The cultures of bacteria and actinomycetes were provided by Drs. F. D. Cook and F. L. Jackson.

Cultures of *Cyathus* were maintained on normal Brodie medium (10). Cultures of other fungi were maintained either on Brodie medium or on Sabouraud's Maltose Agar (Difco). Except for pathogenic forms, all the bacterial and actinomycete cultures were maintained on Difco Nutrient Agar slants. For pathogenic bacteria, a Difco Blood-Agar medium was used. All stock cultures were stored at 5°C and transferred to fresh agar slants at 3- to 4-week intervals. Unless otherwise indicated, all media were autoclaved at 15 lb pressure for 15–20 min at 120°C. The ingredients of the media were either Difco certified products or reagent grade chemicals from Fisher Scientific Company.

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Infrared spectra were recorded on a Perkin-Elmer model 421 dual grating spectrophotometer. Ultraviolet spectra were determined on a Jasco model O.R.D./U.V.-5 spectropolarimeter. Nuclear magnetic resonance spectra were measured on a Varian HA-100 spectrometer, tetramethylsilane being used as an internal standard. In the description of n.m.r. spectra, s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet. Mass spectra including high resolution mass spectra (h.r.m.s.) were recorded on a A.E.I. model MS-9 mass spectrometer. Thin-layer chromatograms (t.l.c.) were carried out using 0.5-mm layers of Silica Gel G (E. Merck AG) containing 1% electronic phosphor; materials were detected either by spraying with 30% sulfuric acid and charring or by viewing under ultraviolet light. Melting points were determined on a Leitz Wetzlar hot-stage melting point apparatus and are uncorrected. Microanalyses were performed by A. B. Gygli, Toronto.

Microbiological Assay of Cyathin

A paper disc – agar plate method for the assay of antibiotic substances was used (11). The test organism was a strain of *Staphylococcus aureus* characterized by its resistance to 10-μg penicillin discs (Difco). Disposable plastic petri plates (100 × 20 mm) were used and 20 ml of Difco antibiotic medium No. 2 was poured into each as a base layer. After solidification, a top layer containing the inoculum was prepared by suspending bacterial cells from a fresh nutrient-agar slant in 5 ml of sterile saline solution and adding this aseptically to 1 liter of the antibiotic medium No. 1 (Difco). A quantity of 5 ml of this inoculum, poured on top of the base layer, provided a uniformly luxuriant growth of bacteria, and clear sharp zones of inhibition resulted. Filter paper discs, 12.7 mm in diameter (Carl Schleicher and Schnell Co., U.S.A.) were used for routine cyathin testing. Discs were dipped in the test solutions, drained free of the excess liquid, and then placed on the bacteria-seeded plate. Plates were incubated at 37.5° and inhibition zones were measured after 18 h of growth.

Assay Procedure for Action Spectrum of Cyathin

For preparing a standard inoculum from the actinomycetes and bacteria, cells from fresh slants were washed, either in sterile distilled water (actinomycetes) or with sterile saline (bacteria). The suspension was adjusted to 20% light transmittance at 600 nm, and 1 ml of this standardized suspension was added to the nutrient-agar medium. Sterile glass petri plates were used and 20–25 ml of seeded medium were poured into each plate. Discs soaked with a known concentration of cyathin were placed on each plate, and inhibition zones were recorded after 24–48 h. The fungal suspension was prepared by washing the spores and (or) mycelium with sterile distilled water and adjusting the transmittance of this suspension to 20% at 560 nm. One milliliter of this suspension was added to 100 ml of the proper medium (Sabouraud's, Czapek's, Brodie's, or malt agar) depending upon the fungi to be tested. Because the growth of some fungi was very slow, a 24- to 48-h period elapsed before the cyathin-bearing discs were placed on the seeded plates. Observations were made after 24–72 h. In any one particular test, the inhibition zones surrounding each of 10 discs were measured.

Instant Thin-Layer Chromatography

Instant thin-layer chromatography was performed using Gelman sheets coated with Silica Gel G of 250-μ thickness. Bioautography was performed using standard bacterial plates seeded with *Staphylococcus aureus* as the test organism. Chromatograms were placed on the assay plates for 30–60 min at 5°C to allow the diffusion of antibiotics in the medium. Plates were then incubated at 37.5 ± 1 for 18 h and the bioactive spots were located.

Large Scale Production, Isolation, and Purification

Cyathus helenae was always grown on 'still-surface' cultures. Brodie's liquid medium containing 20 g dextrose/liter (but no maltose) was used only during the earlier part of this study, as later a chemically defined medium with the following composition was developed: dextrose, 30 g; asparagine, 1.5 g; KH_2PO_4, 1.0 g; $Ca(NO_3)_2 \cdot 4H_2O$, 0.5 g; $MgSO_4 \cdot 7H_2O$, 0.5 g; $ZnSO_4 \cdot 2H_2O$, 250 μg; thiamine hydrochloride, 150 μg; distilled water to make 1 liter.

In either case, a quantity of 200 ml or 400 ml of the medium was poured into a 500-ml or 1000-ml Erlenmeyer flask and the contents autoclaved. The medium was inoculated with three or four 8.0-mm discs of mycelium of *C. helenae* taken from a 16- to 18-day-old colony growing on Brodie's agar medium. Only the growing edges of the colony were used to provide discs of inoculum. The concentrations of cyathin in the culture broth were tested at regular intervals using routine microbiological assay. Maximum yields were achieved between 22 and 25 days at 22–25°C. At this stage, the culture broth was filtered through a thick layer of cheesecloth and the clear brownish liquid was then extracted with an equal volume of ethyl acetate. Several other organic solvents were also tested as extraction media but none proved as effective as ethyl acetate. The resulting yellow-brown organic phase was reduced to dryness under vacuum at 35°C. This operation produced a reddish-brown solid which was readily soluble in acetone, methanol, and ethyl acetate but sparingly soluble in chloroform, ether, and water. In a typical experiment 13.2 g of crude cyathin complex was obtained from 24 l of filtered broth.

Separation of the Cyathin Complex

2,4,5-Trihydroxybenzaldehyde

Crude cyathin complex (1.7 g) was triturated with chloroform (15 ml). The undissolved material (0.59 g) was obtained as a microcrystalline yellowish-brown solid after filtration. Sublimation (140–170°/0.3 mm) followed by recrystallization from chloroform–methanol (9:1) gave yellow crystals of 2,4,5-trihydroxybenzaldehyde, m.p. 216–219° (dec.). Ultraviolet spectrum (methanol): λ_{max} 241 (ε 14 000), 282 (ε 11 000), 350 (ε 8000) nm. Infrared spectrum ($CHCl_3$): 3380, 3160, 1630, 1610, 1590 cm^{-1}. Nuclear magnetic resonance spectrum (acetone-d_6): τ0.25 (1 H, s, —C*H*O), 2.87 (1 H, s, aromatic proton), 3.57 (1 H, s, aromatic proton). Calculated mol. wt. for $C_7H_6O_4$: 154. Found, mol. wt. 154. Exact mass determination was carried out on the *m/e* 153 peak. Calcd. for $C_7H_5O_4$: 153.0188. Found: 153.0187.

This compound was found to be identical (m.p., mixed m.p., infrared spectrum) with authentic 2,4,5-trihydroxybenzaldehyde, synthesized in the following manner (12,

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13): hydroxyhydroquinone (6.3 g) was dissolved in ether (100 ml) and potassium chloride (0.1 g) and zinc cyanide (11.7 g) added. The suspension was cooled in ice and vigorously stirred while a rapid stream of dry hydrogen chloride was passed through the mixture for 1 h. The ether was then decanted and ice (50 g) was added to the residue. The resulting mixture was heated on the steam bath, then cooled and the reddish precipitate (3.0 g) removed by filtration. Sublimation of a portion of this material followed by crystallization from chloroform–methanol gave 2,4,5-trihydroxybenzaldehyde, m.p. 216–219° (dec.), reported 223° (12).

Preparative Thin-Layer Chromatography (p.t.l.c.): Isolation of Cyathin C_5 and Cyathin A_4

A solution of the crude cyathin complex (0.7 g) in ethyl acetate (5 ml) was applied to a 1-meter preparative t.l.c. plate. The plate was developed with benzene : dioxane : acetic acid (100:25:1). Under ultraviolet light, bands were detected at R_f 0.54 (cyathin C_5), 0.51 (2,4,5-trihydroxybenzaldehyde), 0.27 (cyathin A_3), and 0.06 (cyathin A_4).

The material with R_f 0.54 was eluted from the adsorbent with ether–methanol (10:1) to give a brown oil (0.22 g). Further purification by p.t.l.c. yielded a material which showed a single spot on t.l.c. and which crystallized from ether–pentane to give colorless crystals of *cyathin C_5* (40 mg), m.p. 225–226°. Infrared spectrum (Nujol): 3400, 3200, 1695, 1675, 1235, 1220 cm^{-1}. Nuclear magnetic resonance spectrum (acetone-d_6): τ0.1 (1 H, s), 5.16 (2 H, m), 5.9–6.6 (5–7 H, m), 8.92 (3 H, d, splitting 7 Hz), 9.05 (3 H, s), 9.15 (3 H, s).

Anal. Calcd. for $C_{20}H_{26}O_5$ (mol. wt. 346.1780): C, 69.34; H, 7.57; O, 23.09. Found (mol. wt. 346.1774 (h.r.m.s.)): C, 68.85; H, 7.66; O, 23.47.

Cyathin A_4 is most conveniently isolated by subjecting crude cyathin from which most of the trihydroxybenzaldehyde has been removed (by treatment with chloroform, see above) to p.t.l.c. using benzene : acetone : acetic acid (75:25:1) as the developing system. Under these conditions cyathin A_4 has R_f 0.10. In this way, p.t.l.c. of 1.3 g of crude material yielded 0.35 g of a brown oil, which on further chromatography gave *cyathin A_4* (0.25 g) as a white foam, homogeneous on t.l.c. Evaporative distillation at 220°/0.2 mm gave a clear oil which was used for analytical procedures. Infrared spectrum ($CHCl_3$): 3600, 3400, 1700(w), 1650(m), 1375 cm^{-1}. Nuclear magnetic resonance spectrum ($CDCl_3$): τ4.02 (1 H, narrow m), 5.6 (1 H, m), 5.79 (2 H, narrow m), 6.0 (3 H, disappears on D_2O exchange), 6.62 (2 H, m), 9.02 (3 H, s), 9.04 (3 H, s), 9.07 (3 H, d, splitting 6 Hz).

Anal. Calcd. for $C_{20}H_{30}O_4$ (mol. wt. 334.2144); C, 71.82; H, 9.04. Found (mol. wt. 334.2146 (h.r.m.s.)): C, 72.06; H, 8.53.

Cyathin A_4 was further characterized as its crystalline tri-*p*-bromobenzoate prepared in the following way: a solution of cyathin A_4 (40 mg) and *p*-bromobenzoyl chloride (125 mg) in pyridine (2 ml) was kept at room temperature for 3 days, then diluted with water and extracted with ether. The ether extracts were washed successively with dilute hydrochloric acid, saturated sodium bicarbonate solution, and water, then dried and the solvent evaporated. The residue was purified by p.t.l.c., developing being done with benzene–acetone (49:1). Crystallization from methylene chloride–methanol gave *tri-p-bromobenzoylcyathin A_4*, m.p. 188–189°. Infrared spectrum ($CHCl_3$): 1715, 1680, 1580, 1480 cm^{-1}. Nuclear magnetic resonance spectrum ($CDCl_3$): τ2.0–2.5 (12 H, aromatic protons), 3.70 (1 H, narrow m), 4.22 (1 H, t), 5.04 (2 H, m), 5.80 (2 H, AB part of ABX system), 6.72 (1 H, q), 7.0–8.0 (several multiplets), 8.0–8.7 (complex), 8.84 (3 H, s), 8.96 (3 H, d, splitting 7 Hz), 9.02 (3 H, s).

Anal. Calcd. for $C_{41}H_{39}O_7Br_3$: C, 55.72; H, 4.45; O, 12.67; Br, 27.16. Found: C, 56.00; H, 4.49; O, 12.72; Br, 27.25.

Cyathin A_4 gave a noncrystalline triacetyl derivative when treated with acetic anhydride – pyridine and a noncrystalline acetonide when treated with acetone – perchloric acid.

Column Chromatography: Isolation of Cyathin B_3, Allocyathin A_4, Cyathin A_3, and Cyathin B_4

Crude cyathin (1.1 g) from which most of the trihydroxybenzaldehyde had been removed was subjected to chromatography on silicic acid (50 g, 100 mesh, Mallinckrodt reagent grade). The column was made up in chloroform containing 0.75% ethanol, hereafter referred to as chloroform. Elution with chloroform (300 ml) yielded traces of impurities. Elution with a further 130 ml of chloroform gave fraction 1 (36 mg). Elution with 300 ml chloroform–methanol (49:1) gave fraction 2 (0.10 g). Fraction 3 (0.25 g) was eluted with 140 ml of chloroform–methanol (49:1) and fraction 4 (0.19 g) was eluted with 180 ml of chloroform–methanol (19:1). Fraction 5 (0.44 g) was eluted with 300 ml of chloroform–methanol (19:1).

Fraction 1 showed one major spot on t.l.c., and was rechromatographed on silicic acid to give *cyathin B_3* (21 mg), a low-melting solid purified by molecular distillation (100–110°/0.05 mm). Infrared spectrum ($CHCl_3$): 3590, 1680 cm^{-1}. High resolution mass spectrometry indicated the molecular formula $C_{20}H_{28}O_3$ (calcd. mol. wt. 316.2039; found, 316.2038).

Fraction 2 showed several spots on t.l.c. On one occasion this fraction partially crystallized. The crystals were separated and purified by t.l.c., m.p. 85–90°. High resolution mass spectrometry indicated the molecular formula $C_{20}H_{30}O_4$ (calcd. mol. wt. 334.2144; found, 334.2139). Since this compound shows a much higher R_f than cyathin A_4 it appears to be an isomer of A_4 and is tentatively named *allocyathin A_4*. Infrared spectrum ($CHCl_3$): 3600, 3490, 1710 (broad) cm^{-1}.

Fraction 3 contained 2,4,5-trihydroxybenzaldehyde, as indicated by its yellow color and t.l.c. behavior, which was removed in the following manner. Fraction 3, combined with similar material from other chromatograms (total, 0.8 g), was dissolved in ether (100 ml) and extracted with ice-cold 5% aqueous sodium hydroxide. The aqueous layer was then washed three times with ether. The combined ether solutions were dried and evaporated to give *cyathin A_3* (0.31 g) as an oil, homogeneous on t.l.c. The material was purified by evaporative distillation at 110°/0.05 mm. Infrared spectrum ($CHCl_3$): 3580, 3400, 1710, 1650 cm^{-1}.

Anal. Calcd. mol. wt. for $C_{20}H_{30}O_3$: 318.2195. Found: 318.2195 (h.r.m.s.).

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TABLE 1

Antimicrobial action spectrum of cyathin*

Organisms tested	Inhibition zone, mm†
Actinomycetales	
4463 (unidentified)	14.0
4471 (unidentified)	13.0
Unidentified	12.0
J_2 (unidentified)	22.0
Mycobacterium rhodoerinchii	25.0
Streptomyces griseus	19.0
Eubacteriales	
Aerobacter aerogenes	2.0
Agrobacterium tumefaciens	31.0
Bacillus pumilis	5.0
B. subtilis	14.0
Escherichia coli	2.0
Lactobacillus casei	10.0
Micrococcus aurantiacus	14.0
M. candidus	15.0
M. conglomeratus	13.0
M. cyaneus	12.0
M. denitrificans	12.0
M. flavus	12.0
M. freudenerichii	9.0
M. luteus	11.0
M. roseus	14.0
M. varians	12.0
Pediococcus cerevisae	7.0
Proteus mirabilis	3.0
Pseudomonas denitrificans	14.0
Serratia sp.	5.0
Streptococcus faecalis	8.0
Human pathogens‡	
Clostridium welchii	++
Corynebacterium diphtheriae	++
Diplococcus pneumoniae	+
Escherichia coli	−
Haemophilus parainfluenzae	+
Neisseria meningitidis	+
Pneumococcus multocida	+++
Proteus sp.	−
Pseudomonas sp.	−
Salmonella enteritidis	−
S. thompsoni	−
Shigella sonnei	−
Staphylococcus aureus (penicillin resistant)	+++
Streptococcus sp.	−
S. pyogenes	+
Fungi	
Aspergillus amstelodami	3.0
A. leporis	++
A. nidulans	+
A. niger	3.0
Candida krusei	+
Chaetomium sp.	16.0
Dipodascus sp.	2.0
Fusarium solani	8.0
Gliocladium aureus	7.0
Mucor genevensis	+
Penicillium claviforme	++
P. commune	++
P. digitatum	7.0

TABLE 1 (*Concluded*)

Organisms tested	Inhibition zone, mm†
P. ducklauxii	+++
P. expansum	+
P. lilacianum	4.0
P. notatum	6.0
P. raistrickii series	+
P. restrictum	5.0
P. roquefortii	−
Phoma sp.	7.0
Rhizopus sp.	+
Schizosaccharomyces octosporus	2.0
Rhodotorula sp.	+
Torulopsis sp.	+
Trichoderma	7.0
T. lignorum	6.0
Ustilago hordei	+
Zygosaccharomyces daperi	+
Dermatophytes	
Arthroderma uncinatum	+
Microsporum canis	9.0
M. cookii	5.0
Trichophyton mentagrophytes	4.0
T. rubrum	7.0
T. terrestre	9.0

*Average of 10 observations.
†Cyathin concentration 480 μg per disc.
‡Tests performed at Provincial Laboratory; courtesy of Dr. F. L. Jackson.
NOTE: − No inhibition.
+ Zones not well defined; ++, +++: comparatively marked inhibition.

Cyathin A_3 was further characterized as its diacetyl derivative, prepared as follows: cyathin A_3 (50 mg) was kept in pyridine (2 ml) containing acetic anhydride (1 ml) for 2 h at room temperature, then worked up in the usual way. The crude product was chromatographed over silicic acid using alcohol-free chloroform as eluant to give *diacetylcyathin A_3* (27 mg) as a thick oil which was further purified by evaporative distillation (100°/0.05 mm). Infrared spectrum ($CHCl_3$): 1740 (strong), 1680 (broad), 1225 (strong). Nuclear magnetic resonance spectrum ($CDCl_3$): τ3.88 (1 H, narrow m), 4.64 (2 H, m), 5.33 (2 H, AB part of ABX system, X located at 3.88 as shown by double irradiation), 7.0–7.9 (3–4 H, complex series of m), 7.96 (3 H, s), 8.04 (3 H, s), 8.98 (3 H, s), 9.02 (3 H, s), 9.05 (3 H, d, splitting 6 Hz). Ultraviolet spectrum (MeOH): λ_{max} 225 nm (ε 7000).

Anal. Calcd. mol. wt. for $C_{24}H_{34}O_5$, 402.2406. Found: mol. wt. 402.2400 (h.r.m.s.).

Cyathin B_4

During one separation, fraction 3 was found to contain another component of higher R_f value. This was separated by p.t.l.c. and obtained, after molecular distillation, as a colorless viscous oil. High resolution mass spectrometry indicated the molecular formula $C_{20}H_{28}O_4$ (calcd. 332.1988; found, 332.1985). This compound, which has different t.l.c. behavior from the other cyathins reported has been tentatively named cyathin B_4. Infrared spectrum ($CHCl_3$): 3570, 1675, 1595 cm^{-1}.

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Results and Discussion

Antimicrobial Action Spectrum

The results of the activity of cyathin against actinomycetales, bacteria, and fungi are presented in Table I. Amongst the actinomycetales, cyathin was most active against *Mycobacterium rhodoerinchii*.

Gram-positive as well as gram-negative bacteria were sensitive to cyathin, but the former were the more sensitive. All species of *Micrococcus* tested proved to be sensitive. *Staphylococcus aureus* 'penicillin-sensitive' as well as 'penicillin-resistant' strains were found to be sensitive. However, *E. coli* showed little sensitivity. This was also true for *Serratia* sp. and *Proteus mirabilis*. Among pathogenic forms, only *Pneumococcus multocida*, *Corynebacterium diphtheriae*, *Clostridium welchii*, *Diplococcus pneumoniae*, *Streptococcus pyogenes*, *Neisseria meningitidis*, and *Haemophilus parainfluenzae* were sensitive. None of these pathogenic forms, however, exhibited large zones of inhibition. On the other hand, *Agrobacterium tumefaciens* (the cause of crown gall disease of many horticultural plants) was found to be the most sensitive of all the organisms screened. Although no further study of *Agrobacterium* was made, it should be undertaken since crown gall is a serious plant disease, frequently of economic importance.

All the fungi tested against cyathin were sensitive to some degree. Most of the sensitive ones belong to the class Deuteromycetes. The most striking examples of fungi sensitive to cyathin include species of *Aspergillus*, *Penicillium*, *Fusarium*, *Trichoderma*, *Chaetomium*, and *Gliocladium*. However, none of the Phycomycetes proved to be sensitive to cyathin. The same was true of yeasts.

Of the six 'dermatophytes' tested, all showed some degree of sensitivity to cyathin. The most striking examples were *Microsporum canis*, *Trichophyton rubrum*, and *T. terrestre*.

TABLE 2

Instant thin-layer chromatography of cyathin

Groups	Solvent system	R_f
α	Methanol	0.4–0.8
β	10% methanol in chloroform	0.0–0.28 0.29–0.75
γ	Chloroform	0.0–0.08 0.14 0.48
I*a*	Pyridine:water (1:1)	0.4 0.61
I*b*	Pyridine:water:ethanol (1:1:1)	0.44 0.48–0.72
I*c*	Pyridine:water:ethanol (1:1:3)	0.4–0.64
II*a*	Butanol:methanol	0.72
II*b*	Chloroform:methanol	0.39 0.49–0.8
II*c*	Absolute ethanol	0.49–0.74
III*a*	Methanol:benzene (12:88)	0.38 0.64
III*b*	Methanol:benzene (6:94)	0.0
III*c*	Methanol:benzene (4:96)	0.0
IV*a*	Methanol:benzene (1:99)	0.0
IV*b*	Methanol:benzene:chloroform (1:49:50)	0.0

TABLE 3

Comparison of cyathin with known antibiotics of subgroup IV-I

	Solvent system				
Antibiotic	α	β	γ	IV*a*	IV*b*
Actinibolin	0.42	0.2	0.08	0	0
DON	0.78	0.76	0.65	0	0
Psicofuranine	0.7	0.15	0–0.1	0	0
Spiramycin	0.6	0.4	0–0.3	0	0
Cyathin	0.4–0.8	0–0.28 0.29–0.75	0.08 0.14 0.48	0	0

Instant Thin-Layer Chromatography of Cyathin

It is common experience in research on antibiotics to find that some supposedly new antibiotic turns out to be a known one. To avoid this problem, an effort was made at an early stage to assess the probability that the antibiotic in question does not fall into this class. An instant thin-layer chromatography method, devised by Aszalos *et al.* (14), proved to be useful. This method makes use of the movement of an antibiotic in a specific solvent system. The first three solvent systems α, β, and γ (Table 2) divide substances into four primary groups; these are then subdivided on the basis of the use of 11 additional solvent systems. As may be seen from Table 2, cyathin exhibited movement in all the first three solvent systems (α, β, and γ). Of the additional 11 solvent systems, cyathin did not exhibit any

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movement in solvents III*b*, III*c*, IV*a*, and IV*b*. This places cyathin in subgroup IV-I of the scheme of Aszalos *et al.* This subgroup contains only four antibiotics, viz. actinobolin, DON, psicofuranine, and spiramycin. Table 3 shows the R_f value of these four antibiotics and of cyathin in solvents α, β, γ, IV*a*, and IV*b*. It may be noted that, in terms of R_f values, cyathin did not match any of the four known antibiotics in all the solvents. To study the ionic behavior of cyathin, chromatography was performed on buffered silica-gel sheets impregnated with 0.5 *M* KH_2PO_4 buffer solution of pH 2 or pH 11. The chromatograms were developed using the least polar solvents in which cyathin showed movement (refer to Table 2). The results are expressed in Table 4.

TABLE 4
Buffered instant thin-layer chromatography of cyathin

Solvent system	R_f pH 2.0	R_f pH 11.0
Chloroform	0–0.29 and 0.31–0.64	0
Methanol	0.39 and 0.78	0.34–0.99

TABLE 5
Yield of cyathin in batch culture

Harvest	Medium	Volume of culture fluid, liters	Yield, g
1	Undefined*	17	5.76
2	Defined†	11	2.54
3	Undefined	15.5	3.54
4	Defined‡	6	1.0

*Dextrose 20 g/liter in Brodie medium (no maltose).
†Chemically defined medium (dextrose 30 g/liter).
‡Chemically defined medium (maltose 30 g/liter; no dextrose).

TABLE 6
Thin-layer chromatography behavior (R_f values) of cyathin components

Compounds	R_f values 1*	2*	3*
Cyathin A_4	0.09	0.06	—†
Cyathin A_3	0.34	0.27	—
2,4,5-Trihydroxybenzaldehyde	0.44	0.51	0.44
Cyathin B_4	0.50	—	—
Cyathin C_5	0.54	0.54	0.55
Cyathin B_3	0.56	—	—
Unidentified	—	0.63	0.60
Unidentified	—	0.67	0.67

*Solvent systems: 1, benzene – acetone – acetic acid (70:30:1); 2, benzene – acetone – acetic acid (72:25:1); 3, benzene – dioxane – acetic acid (100:25:1).
†A dash indicates that the compound was not detected or was not positively identified.

The yields of cyathin in separate batch cultures varied to some extent. It was realized during the latter part of this study that the fungus *C. helenae* has a tendency to produce sectors quite frequently and because these sectors are unstable there was always a variation in the titers of cyathin. Table 5 gives an example of the amounts of crude cyathin obtained from four different harvests.

The earlier i.t.l.c. studies indicated that cyathin could be a new antibiotic, but whether or not it was a single active principle remained to be explored. As discussed below, cyathin is actually a mixture containing several components. When these components were tested for biological activity against *S. aureus*, almost all of them showed antimicrobial activity.

Chemistry of the Cyathins

The active constituents of the culture broth were readily extracted with ethyl acetate. Examination of the crude extract (referred to as cyathin) by t.l.c. revealed the presence of at least eight components (see Table 6). One of the components was sparingly soluble in chloroform and was separated by triturating the crude cyathin with chloroform. The insoluble component was a yellow crystalline material with a very distinctive ultraviolet spectrum. It possesses the molecular formula $C_7H_6O_4$ and its spectral characteristics and melting point (12) suggested that it is 2,4,5-trihydroxybenzaldehyde. This was confirmed by synthesis of an authentic sample. It appears that 2,4,5-trihydroxybenzaldehyde has not previously been obtained from natural sources.

Several other components of cyathin have been isolated by means of preparative thin-layer chromatography on silica gel and (or) elution chromatography over silicic acid. These all have proved to be oxygenated C_{20} compounds having either 26, 28, or 30 hydrogens. The following system of trivial nomenclature has been adopted. Compounds containing 30 hydrogens are called cyathin A; 28 hydrogens, cyathin B; and 26 hydrogens, cyathin C. The subscript after the letter denotes the number of oxygens, e.g. cyathin C_5 is $C_{20}H_{26}O_5$. Isomers of the first isolated cyathins are placed in the allo series, e.g. both cyathin A_4 and allocyathin A_4 are $C_{20}H_{30}O_4$.

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The most abundant cyathins are cyathin A_4 and cyathin A_3. The structures of the cyathins remain to be determined, but preliminary evidence suggests that they may represent a new group of diterpenoids.

To date, the chemistry of cyathin A_4 ($C_{20}H_{30}O_4$) has been examined in greatest detail. Three of the oxygen functions are present as hydroxyl groups as revealed by the formation of triacetyl and tri-*p*-bromobenzoyl derivatives. These derivatives do not show hydroxyl absorption in the infrared, and since cyathin A_4 does not show distinctive carbonyl absorption, it appears that the fourth oxygen is present either as an ether or as a masked carbonyl. Cyathin A_4 forms an acetonide, thus two of the hydroxyl groups must be in a 1,2- or 1,3-relationship. Nuclear magnetic resonance spectra of cyathin A_4 and its triacetyl and tri-*p*-bromobenzoyl derivatives indicate that two of the alcohol functions are primary and one is secondary. The n.m.r. spectrum also shows an olefinic proton (catalytic hydrogenation gives rise to a dihydro derivative) as well as a secondary and two quaternary C-methyl groups. Since only one unsaturated linkage can be detected, it appears that the molecule is pentacyclic.

Examination of the reactions and the spectra of cyathin A_3 and its derivatives indicate that it is closely related to cyathin A_4, the major difference being that one of the primary hydroxyl groups of cyathin A_4 appears as a methyl group in cyathin A_3.

The chemistry of allocyathin, cyathin B_4, cyathin B_3, and cyathin C_5 has not been explored as yet. In some of the preparations of the cyathin complex we have been unable to detect cyathin C_5. The examination of the structures of the cyathins is continuing.

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